

MATHEMATICS 2250-3
Ordinary Differential Equations and Linear Algebra
Fall semester 1999

texts:

Differential Equations,
Computing Projects for Differential Equations supplement,
and *Elementary Linear Algebra* Chapters 1-4
by C.H. Edwards Jr. and David E. Penney

when: M, W, F 10:45-11:35 A.M.

where: NS 204

instructor:	Prof. Christopher Hacon
office:	JWB 118
telephone:	581-8341
email:	hacon@math.utah.edu
office hours:	M 9:45-10:35, W 9:45-10:35, F 12:40-1:20.

course outline: This course is an introduction to differential equations, and how they are used to model problems arising in engineering and science. Linear algebra is introduced as a tool for analyzing systems of differential equations, as well as standard linear equations. Particular attention is paid throughout the course to qualitative behavior of solutions, stability, and numerical approaches. Computer projects will be assigned to enhance the material.

We will start by quickly covering most of Chapters 1 and 2 in Edwards and Penney on linear differential equations. Then we will study linear algebra for four weeks covering the four chapters of the supplemental text. Then we will return to differential equations and study higher order equations (Chapter 3), systems of differential equations (Chapters 4 and 5) and end with Chapter 7 studying Laplace Transforms.

There will be four computer projects assigned during the semester, related to the classroom material. They will be written in the software package MAPLE. There is a Math Department Computer Lab at which you all automatically have accounts, and there are other labs around campus where Maple is also available, for example at the College of Engineering. You are responsible for learning to use a computer system and the programs needed to complete the assignments. Introductory lectures on the computer lab and Maple will be given in the first three weeks of the semester. There will be a tutoring center support for these projects (and for your other homework) as well. The Math Department Lab and Engineering Math Tutoring Center are both located in Building 129, between JWB and LCB.

You will be permitted to work your solutions to the computer projects in different languages, such as Mathematica or Matlab, if you prefer these, although it will be easier to find help if you are using Maple. Notice that the text and the *Computing Projects* supplement support all three of these languages.

grading: There will be three midterms, a comprehensive final examination, and homework. Each midterm will count for 20% of your grade (only the highest two midterm grades will be considered, you will be allowed to drop your lowest score, however there will be no make up tests), homework will count for 15% (you will be allowed to drop your lowest 3 homework scores, however late homework will not be accepted), maple assignments will count for 15%, and the final exam will make up the remaining 30%. The book homework will be assigned and collected weekly, on Fridays. Maple project homework will generally be due one week after it is assigned. A homework grader will partially grade your assignments. The value of carefully working homework problems is that mathematics (like anything) must be practiced and experienced to be learned.

It is the Math Department policy, and mine as well, to grant any withdrawal request until the University deadline of October 22 (there is no tuition penalty if dropped by September 3rd).

Final: Wed. Dec. 15 from 9:15 to 11:15.

Tutoring and help: The Math Department's tutoring center is available for help with homework from approximately 8 a.m. to 8 p.m. M-Th and 8 a.m. to 2 p.m. on Friday. Additionally the computer lab is staffed with assistants and is open M-Th 8am-10pm, Friday 8am-6pm and 10am-5pm on the weekends. The computer lab assistants are provided to help with general computing questions and basic Maple questions but they may not be able to offer specific help on the mathematics of this course. There is a "quiet" study room in bldg. 122 across from the Undergraduate commons.

Homepage: You may refer to the course homepage for further information and future handouts.
<http://www.math.utah.edu/~hacon/math2250.html>

ADA statement: The American with Disabilities Act requires that reasonable accommodations be provided for students with physical, sensory, cognitive, systemic, learning, and psychiatric disabilities. Please contact me at the beginning of the semester to discuss any such accommodations for the course.